When Static Thresholds Fail: Online Conformal Calibration for Cross-Domain Industrial Anomaly Detection

Hu Lingyan and Zhu Jibiao

***Abstract*—Industrial anomaly detectors are typically trained and calibrated on one operating domain, then deployed on another. We study what happens to the deployed *decision threshold* under this cross-domain shift. Using six leave-one-group-out folds spanning semiconductor fabrication, wind, solar, and rotating-machinery monitoring at native sampling rates, we show that a static source-calibrated threshold fails in both directions at a nominal 0.90 normal-window coverage level: coverage ranges from 0.18 (severe over-alarming) to 0.999 (no alarms). An online adaptive conformal inference (ACI) layer restores coverage to within 0.015 of nominal on five of six folds across four nominal levels, with bootstrap confidence intervals for the coverage gap of at most 0.02; rolling-quantile and extreme-value baselines do not recover coverage, and the quantity ACI is entitled to regulate—the stream-wide alarm rate—tracks the nominal level on every fold (largest deviation 0.033). We then identify an *alarm-budget defect*: ACI regulates the stream-level alarm rate, so on folds where anomalies are abundant and separable the budget is consumed by anomalies and normal-window coverage over-shoots. A normal-anchored variant (NA-ACI) that updates the quantile only on likely-normal windows removes the defect—detection rate on the machinery fold rises from 0.12 to 0.90 (at κ=3) with essentially unchanged normal-window FPR ($0.090 \to 0.094$), and mean range-F_1 across folds improves from 0.26 to 0.39—while matching ACI on anomaly-sparse folds. Conditional (Mondrian) and divergence-shifted variants otherwise bring no measurable coverage gain, and a band-spectral divergence estimate does not predict the measured transfer cost (threshold inflation $0.5\times$–$19\times$). Calibration and detection quality decouple: on two folds with below-chance score rankings, coverage remains controllable but alarms carry no detection utility—a caveat we state explicitly rather than hide behind the coverage numbers. We release the benchmark, labeling and injection protocols, and code, along with a deployment guideline.**

***Index Terms*—Anomaly detection, conformal prediction, distribution shift, industrial time series, calibration, reliability.**

I. Introduction

Industrial monitoring pipelines rarely stop at producing an anomaly score. Operators need a decision: when is a score high enough to trigger an inspection or a maintenance action? That decision requires a *threshold*, which must be set without target-domain labels. The standard practice is to calibrate the threshold on source data and reuse it after deployment. We ask a simple question that, to our knowledge, has not been answered systematically: what happens to that threshold when the deployment domain differs from the training domain?

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The question matters because cross-domain anomaly detection research has largely optimized the *ranking* quality of detectors—autoencoders, contrastive domain adaptation, or time-series foundation models—while treating the decision threshold as an afterthought. Conformal prediction supplies distribution-free thresholds, but its exchangeability assumption is violated by both temporal dependence and domain shift; the adaptive-conformal literature has built machinery for temporal drift, yet a cross-*domain* (cross-population) study on real industrial data is missing.

This paper reports such a study. We assemble six leave-one-group-out folds (dataset-level groups) over four domain families—semiconductor chemical-mechanical polishing, wind SCADA, photovoltaic plant measurements, and rotating machinery (three datasets)—keep each dataset at its native sampling rate, and fix documented labeling or injection protocols. Two guardrails are pre-committed: folds whose scores rank anomalies below chance are excluded from detection-quality claims, and the synthetic-only fold is flagged in every table. On every fold we train the same plain reconstruction encoder on source-normal windows and then deploy the score without target labels. We compare four thresholding mechanisms: a static source quantile, online adaptive conformal inference (ACI) [1], a band-spectral divergence-shifted variant, and a conditional (Mondrian) variant that adapts separate quantiles across divergence regimes.

The results are unambiguous on the threshold side and honest about the detector side. Static thresholds fail in *both* directions: on the wind fold the source threshold over-alarms so heavily that normal-window coverage collapses to 0.18, while five folds under-alarm with coverage 0.989–0.999 at a nominal 0.90 level. Online ACI restores coverage to within 0.015 of nominal on five folds across four nominal levels, and its advantage is not shared by rolling-quantile tracking or an extreme-value tail fit. Two additional findings cut against our own initial hypotheses and we report them as negative results: a divergence-shifted quantile adds nothing over plain ACI, and a band-spectral divergence statistic does not predict the transfer cost across folds. Finally, calibration and detection quality decouple—on two folds whose scores rank anomalies *below* chance (a property of the datasets and labeling protocols, not of our encoder; feature baselines rank equally badly), ACI still controls coverage, but the alarms are useless for detection. We state plainly that calibration cannot rescue an anti-correlated ranker; coverage control is not a substitute for discriminative scores.

Our contributions are:

- **Benchmark.** A six-fold, four-domain-family leave-one-domain-out benchmark for cross-domain industrial anomaly detection at native sampling rates, with fully documented labeling and synthetic-injection protocols, provenance-checked datasets, and released code and configurations at https://github.com/JibiaoZhu/crossdomain-conformal-calibration (archived at https://doi.org/10.5281/zenodo.23321794).
- **Calibration study.** Evidence that static source thresholds are unsafe under cross-domain shift in both directions, and that an online adaptive-conformal layer restores normal-window coverage on five of six folds, with bootstrap confidence intervals, an update-rate ablation, and calibration baselines (rolling quantile, extreme-value tail, slower updates).
- **Method.** We identify an *alarm-budget defect* of online ACI on anomaly-rich streams—the budget is consumed by anomalies and normal-window coverage overshoots—and propose normal-anchored ACI (NA-ACI), which restricts the quantile feedback to likely-normal windows. NA-ACI reallocates alarms to anomalies (machinery-fold detection $0.12 \rightarrow 0.90$ with essentially unchanged FPR) and raises mean range-F_1 from 0.26 to 0.39 at comparable coverage, with an assumption-based FPR bound and an online exclusion-rate diagnostic; a single parameter interpolates toward ACI on anomaly-sparse folds.
- **Negative results and diagnostics.** Divergence-shifted and conditional (Mondrian) threshold variants do not improve over ACI in this study; band-spectral divergence does not predict threshold inflation; and mechanism-heavy detector components (invariance losses, test-time adaptation, source gating) proved marginal in earlier ablations, which is why this paper studies calibration rather than architecture.
- **Operational analysis.** False-alarm rates, detection delays, threshold traces, transfer-cost inflation ($0.5\times$–$19\times$), and per-run runtime/memory of the calibration layer (0.6–$0.8\,\mu$s per decision, 26 MB peak), supporting a deployment guideline.

Section II positions the work against cross-domain detectors and conformal prediction for time series. Section III defines the problem, metrics, datasets, and protocols. Section IV describes the calibration framework and its variants. Section V presents the main results and analyses. Section VI distills the deployment guideline and limitations, and Section VII concludes.

II. Related Work

A. Cross-Domain Industrial Anomaly Detection

Deep transfer learning for industrial time-series anomaly detection is surveyed in [2], which organizes methods by transfer setting and lists data gaps as a main obstacle. Two recent systems define the state of the art in feature-space transfer: CICADA [3] learns interpretable mixture-of-experts codes that expand to unseen target domains, and DACAD [4] combines anomaly injection with contrastive unsupervised domain adaptation. Both optimize detection scores; neither produces calibrated decision thresholds, and both assume aligned feature spaces that ignore frequency structure. Our study takes their score-producing pipeline as given and asks the complementary reliability question: at what threshold should the score be deployed, and does the answer survive a domain change?

B. Time-Series Foundation Models Under Shift

Chronos [5] and PatchTST-style backbones [6] popularized reusable sequence representations for forecasting, and they are frequently used zero-shot for anomaly scoring. Two recent analyses temper this: [7] isolates *spectral shift*—a mismatch between downstream dominant frequencies and pretraining spectra—as a driver of foundation-model failure, and [8] argues that the diversity of real series undermines the foundation-model premise. Neither proposes a deployable thresholding fix. Our benchmark deliberately keeps native sampling rates so that spectral shift is preserved rather than resampled away.

C. Conformal Prediction for Time Series

Split conformal prediction [9] gives distribution-free marginal coverage from exchangeable calibration data; time series and domain shift violate exchangeability. Adaptive conformal inference (ACI) [1] replaces exchangeability with an online feedback rule on the conformal quantile and guarantees long-run miscoverage under arbitrary dependence, and [10] systematizes update variants (including the DtACI multiplicative-plus-additive rule) for temporal streams. Conformal PID control [11] casts the update as a controller and adds proportional–derivative terms. Extreme-value thresholding [12] is the classical streaming alternative, controlling tail exceedance with a generalized-Pareto fit. All of this work studies temporal drift within a domain; cross-domain (cross-population) calibration on real industrial data, to our knowledge, has not been evaluated systematically. We provide that evaluation, including the cases where the conformal layer succeeds and the cases where it cannot help. Conformal anomaly detection has its own lineage—sequential conformal detectors for streaming monitoring [13] and risk-controlling prediction sets that guarantee downstream decision risks [14]—and our threshold controller sits squarely in it: it is a risk-controlled decision layer placed after a frozen, uncalibrated score. Weighted conformal prediction [15] is the principled answer to covariate shift; our divergence-shifted variant approximates it crudely through a band-spectral signal, and Section V-G reports its honest score-space counterpart, whose gains concentrate on the support-overlapping fold (F2) and vanish under the near-disjoint scale jump of F6. NA-ACI is deliberately *not* weighted conformal: it does not reweight the calibration measure but restricts the feedback path to a sub-stream, which preserves the label-free online update and its alarm-rate semantics (a design related to filtered/robust ACI variants, with an explicit exclusion-rate diagnostic). For deployment-time monitoring, change-point detectors (ADWIN [16]; Page–Hinkley CUSUM

[17]) are the standard recalibration triggers, connected to our guideline in Section VI, and label-shift correction of classifier outputs [18], [19] is the standard tool when prevalence changes—Section V-G reports our attempt to use it for automatic κ selection, and why it fails on these data.

D. Parameter-Efficient Adaptation

Test-time adaptation and parameter-efficient fine-tuning—e.g., the robust frequency-domain adapter of PETSA [20]—are natural candidates for closing cross-domain gaps. In our earlier ablations, test-time adaptation and gating brought no measurable detection gain at this scale, and we therefore keep the encoder fixed and study the calibration layer in isolation; adaptation methods remain compatible with the framework as score providers.

III. Problem Setup and Benchmark

A. Setting and Metrics

A detector produces a window-level anomaly score s_t from a multivariate window x_t (window length 256 samples, stride 128, native sampling rates throughout). Training uses *source-normal* windows only, pooled across source domains. At deployment, the target stream is unlabeled; we may use an initial segment as an unlabeled calibration stream but never its labels. A calibration rule turns scores into thresholds q_t and alarms $\mathbb{1}[s_t > q_t]$.

We report the following quantities. *Normal-window coverage* $\mathrm{cov} = 1 - \mathrm{FPR}$ is the fraction of evaluation windows with label "normal" that are not alarmed, with the requirement $\mathrm{cov} \approx 1 - \alpha$ at nominal level α; unless stated otherwise, coverage is the calibration target and all calibration tables report it. The *coverage gap* is $\Delta = |\mathrm{cov} - (1-\alpha)|$. We also report the *alarm rate* over the whole stream, because an online rule that regulates alarms over a stream containing anomalies cannot in general also control the normal-window FPR: the two coincide only when anomalies are rare relative to α. *Threshold inflation* is $\overline{q_t}/q_{\mathrm{src}}$, the mean deployed threshold relative to the source-calibrated quantile; it measures the transfer cost of keeping coverage. Detection-side metrics are range-based F_1, AUROC/AUPRC, false alarms per window (equivalently per 1000 windows), and detection delay in windows after an anomaly run begins. Window duration is set in samples (256 at stride 128), so its physical span varies with the native rate—from 4.3 min on F1 (1 Hz) to 5–21 ms on the kHz folds; we report physical-time equivalents where absolute timing matters. Notation: cov is normal-window coverage ($1-\mathrm{FPR}$); $1-\alpha$ is the nominal coverage target; *alarm rate* is the fraction of *all* evaluation windows alarmed; *threshold inflation* is the deployed-to-source threshold ratio; every reported quantity names its measurement domain (normal windows vs whole stream). All calibration results are averaged over three training seeds. For uncertainty we use hierarchical moving-block bootstrap: evaluation windows are resampled in contiguous blocks within each fold to respect temporal dependence, and block-length sensitivity (50/200/1,000 windows) is reported.

TABLE I
Folds, datasets, native rates, and evaluation sets (windows of length 256, stride 128). F4's Paderborn component is the time-varying run-to-failure release at 128 kHz; F5's fault sections are sampled at 48.8 kHz, half the 97.7 kHz baseline. Anomaly fractions vary by protocol and matter for interpreting alarm-budget effects.

Fold	Target (rate)	Rate	Normal	Anom.	Label source
F1	PHM16-CMP (1 Hz)	1 Hz	1912	251	removal-rate q_{10} drift
F2	CARE wind (10 min)	10 min	4714	557	status labels
F3	PVDAQ solar (∼10–15 min)	∼12 min	858	214	synthetic injection
F4	CWRU+Paderborn	12/128 kHz	10980	7556	seeded / last-20%
F5	MFPT	97.7/48.8 kHz	5072	13661	segment labels
F6	XJTU-SY	25.6 kHz	12806	3993	last-20% run-to-failure

B. Six Folds over Four Domain Families

We construct six leave-one-group-out folds (each fold holds out one dataset group; four domain families are covered). Table I lists targets, native rates, evaluation set sizes, anomaly fractions, and label provenance; sources for each fold are all other groups' normal windows. Two guardrails apply throughout: folds whose ranking quality is below chance are excluded from detection-quality claims (diagnosed in Section V-J), and the synthetic-injection fold (F3) is flagged in every table and figure caption. Two dataset-engineering decisions follow from pilot experiments and are fixed throughout: (i) windows are standardized with per-dataset global statistics—per-window standardization removes the amplitude component of degradation and we observed it destroy ranking on run-to-failure data; (ii) no resampling to a common rate is performed, so spectral differences between domains are preserved.

C. Labeling and Injection Protocols

Pseudo-labeling is unavoidable in parts of this benchmark, and each protocol is a documented limitation. PHM16-CMP [21] has no fault labels; we mark a wafer-stage run anomalous when its removal rate falls in the lowest per-stage decile (q_{10}), giving an 11.6% anomaly fraction. CARE wind [22] carries status-based anomaly labels that we use directly. PVDAQ [23] contains normal-operation data only; the F3 fold evaluates a synthetic injection protocol—$+6\sigma$ spikes, $+4\sigma$ level shifts, and segment dropouts applied to 20% of evaluation windows with a fixed seed—and is reported as synthetic-only. Dropouts zero a contiguous span (one-eighth to one-third of the window, both channels) in the standardized window and are scored like any other window, with no masking. Band-spectral features partition the normalized frequency axis (0–Nyquist) into eight equal-width rFFT power bins computed on globally standardized windows, so band comparisons are relative-frequency and amplitude-excluded by construction. CWRU supplies seeded-fault files (label normal/fault). Paderborn [24] is the time-varying run-to-failure release (bearings B01–B04 at 128 kHz; ∼1.6 s measurements accumulated over bearing life)—distinct from the classic fixed-condition Paderborn set—and we label the last 20% of measurement indices anomalous as late-life pseudo-labels. MFPT segments carry per-segment labels (baseline segments at 97.7 kHz, fault sections at 48.8 kHz). XJTU-SY [25] follows the last-20% run-to-failure convention. All folds share the same encoder and calibration code; dataset sizes are equalized by deterministic window subsampling

(seed fixed) where needed. Exact file selections are fixed in the released configuration: CWRU records 97–100 (normal) and 105, 106, 107, 108, 118, 130, 169, 185 (faults); Paderborn bearings B01–B04 (40 sampled measurements each, 8 windows per measurement); XJTU-SY bearings 1_1, 1_2, 1_3, 2_1, 3_1; CARE assets 12, 16, 33, 35, 52; and MFPT baseline/fault segments; windowing, labeling, and injection code accompany this paper at https://github.com/JibiaoZhu/crossdomain-conformal-calibration (archived at https://doi.org/10.5281/zenodo.23321794). Throughout, "anomaly" denotes the operational or quality deviation defined by each protocol—a low removal-rate run, a status flag, a late-life window, or an injected perturbation—not necessarily a physical fault; F1's labels in particular mark process outliers rather than hardware failure.

D. Encoder and Training

Scores come from a plain PatchTST-style reconstruction model [6]: patch length 64, stride 32, $d_{\text{model}} = 64$, 4 heads, 2 encoder layers, band-conditioned patch embeddings; the per-window score is the mean squared reconstruction error. Models train for 4000 steps with batch size 64 (AdamW, learning rate 5×10^{-4}, cosine schedule, AMP) on pooled source-normal windows; target scores are produced zero-shot. The step budget is fixed across folds and seeds with no validation-based early stopping, keeping encoders identical across folds; the terminal logged reconstruction loss is ≈ 0.015 (mean over the final interval, no NaNs; per-fold logs released). No test-time adaptation, gating, or invariance regularizers are used in the main runs: earlier ablations showed their contributions were marginal ($\leq$ 2.2% range-F_1 changes across all four Stage-0 folds) and they are reported as negative controls in Section V-N.

IV. Calibration Framework

A. From Scores to Thresholds

A score alone does not trigger maintenance; a threshold does. We compare four thresholding mechanisms on identical scores. All mechanisms estimate or maintain a quantile of the score distribution and alarm when $s_t > q_t$. Only the static rule uses no target information beyond what is deployed at time zero.

B. Static Source Calibration

Given source-calibrated scores (a held-out 20% of source-normal windows), the split-conformal threshold is the empirical $(1-\alpha)$ quantile q_{src}. This is the standard practice implied by [9] and is optimal under exchangeability between calibration and deployment windows—precisely what cross-domain shift breaks.

C. Online Adaptive Conformal Inference (ACI)

We maintain q_t with the *relative-gain* (scale-free) form of adaptive conformal inference [1]

$$q_{t+1} = q_t + \gamma \cdot \max(|q_t|, \epsilon)\,\big(\mathbb{1}[s_t > q_t] - \alpha\big), \qquad (1)$$

a relative-gain integral controller on the alarm indicator: if alarms exceed the target rate the threshold rises, and vice versa. The published ACI update is additive with fixed step [1]; we adopt the scale-free form because the score scales of our folds span five orders of magnitude, so no single additive step serves them all. On the log-threshold $z_t = \ln q_t$ the update is exactly two-step additive with steps $a_+ = \ln(1+\gamma(1-\alpha))$ and $a_- = -\ln(1-\gamma\alpha)$, both $O(\gamma)$; the standard two-point stochastic-approximation argument then places the long-run alarm rate at the equilibrium frequency $a_-/(a_+ + a_-) = \alpha + O(\gamma)$, provided the threshold stays bounded away from 0 and ∞ (the floor $\epsilon = 10^{-6}$ and finite scores ensure this). The initial value is the source-calibrated quantile, $q_0 = q_{\text{src}}$, at the same nominal level. We therefore interpret our results under an $O(\gamma)$ long-run alarm-rate deviation, validate that target empirically (Table VII), and do not inherit the finite-sample constants proved for the additive parameterization. We use $\gamma = 0.05$ and update every window (the update-rate ablation in Section V-E); the DtACI-style multiplicative-plus-additive alternative [10] is bracketed by our slower/larger-gain ablation variants, and conformal PID control [11] differs from Equation 1 in additional proportional-derivative terms, which our ablations found unnecessary at this scale—although its anti-windup motivation lines up with the oscillation pathology documented in Section V-C.

D. Normal-Anchored ACI

ACI's guarantee is on the stream-wide alarm rate. When anomalies are abundant and separable—up to 73% of windows in our machinery folds—an ACI layer spends its budget on anomalies and the normal-window coverage over-shoots (the alarm-budget defect of Section V-C). We propose *normal-anchored ACI* (NA-ACI), which performs the update of Equation 1 only on windows whose score is not far above the current decision threshold: with $u_t = \kappa\, q_t$,

$$q_{t+1} = \begin{cases} q_t + \gamma\,|q_t|\,\big(\mathbb{1}[s_t > q_t] - \alpha\big), & s_t \leq u_t, \\ q_t, & s_t > u_t, \end{cases} \qquad (2)$$

with $\kappa \geq 1$. Alarms still fire for all windows above q_t; only the feedback is restricted. Windows above u_t are treated as likely anomalies and excluded from steering the quantile, so the budget is spent on the normal regime. Intuitively, NA-ACI is ACI run on the retained (predominantly normal) sub-stream: if excluded windows contain at most a fraction ϵ of normal windows, the long-run alarm rate on the retained stream is α, giving a normal-window FPR bound of $\alpha/(1-\epsilon)$ up to finite-sample error—a proposition under the stated assumption, not a distribution-free guarantee, and the assumption itself (anomaly-score separation in the tail of the retained stream) is testable online by monitoring the exclusion rate. Small κ targets the budget aggressively but risks excluding normals during scale drift, which over-alarms; large κ recovers ACI. We evaluate $\kappa \in \{3, 5\}$ alongside the ACI and divergence/Mondrian variants.

Proposition 1 (NA-ACI FPR bound)**.** *Assume (A1) the retained stream ($s_t \leq \kappa q_t$) contains anomalous windows at*

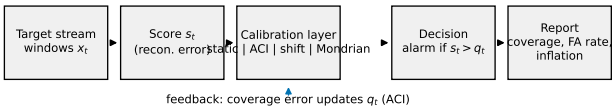


Fig. 1. Calibration framework. A score producer (plain reconstruction encoder, frozen) feeds a calibration layer that estimates or tracks a threshold: static source quantile, online ACI with coverage feedback, divergence-shifted, or conditional (Mondrian). Deployments are summarized by normal-window coverage, alarm rate, false alarms, detection delay, and interval inflation.

fraction at most ϵ, and (A2) the update of Equation 2 runs with the scale-free gain of Section IV with floor ϵ_q and bounded scores. Then NA-ACI's long-run alarm rate on the retained stream equals $\alpha + O(\gamma)$, and its false-alarm rate among retained normal windows is at most $\alpha/(1-\epsilon) + O(\gamma)$.

Proof sketch. (i) The excluded windows are removed by a causal rule that depends only on past thresholds, so the retained subsequence preserves order; on it, the update is the same two-point stochastic approximation analyzed in the remark following Equation 1, and its equilibrium argument gives alarm rate $\alpha + O(\gamma)$ on the retained stream. (ii) Among the retained windows, the number of alarms is $\alpha|R|(1+o(1))$, and (A1) gives at least $(1-\epsilon)|R|$ retained normals, so their alarm rate is at most $\alpha/(1-\epsilon) + O(\gamma)$. (iii) Since $\kappa \geq 1$, every excluded window has $s_t > \kappa q_t \geq q_t$ and is alarmed by construction; the exclusion rate—monitored online, as in Section V-C—therefore bounds the additional share of alarmed normals outside the retained stream. Assumption (A1) is not distribution-free; it is a testable operating condition, satisfied when the score tail that triggers exclusion is anomaly-dominated, and monitored via the exclusion rate. □

E. Divergence-Shifted and Conditional Variants

Two target-aware refinements are evaluated as hypotheses in their own right. *Divergence-shifted calibration* scales the threshold state by an estimated shift signal: $q_t = q_t^{\text{state}}(1 + \lambda D_t^{\text{band}})$, where D_t^{band} is a per-window band-spectral divergence—the Mahalanobis distance of the window's per-band log-energy vector from source-normal band statistics, recentered and rescaled by the target stream's own early quantiles—and $\lambda = 0.1$. *Mondrian ACI* bins windows into divergence tertiles and runs an independent ACI quantile per bin, targeting conditional coverage across shift regimes; the divergence runs are evidence for or against conditioning signals beyond plain ACI.

F. Transfer-Cost and Diagnostic Measures

Beyond coverage we record *threshold inflation* $\overline{q_t}/q_{\text{src}}$ per fold, and the cross-fold Pearson correlation between the mean divergence $\overline{D}^{\text{band}}$ and inflation. The latter operationalizes the hypothesis that divergence estimates *predict* the price of transfer; Section V-I reports it as a null result with its sample size. Figure 1 summarizes the pipeline.

V. Experiments

A. Setup Recap

Each fold trains one encoder on pooled source-normal windows (3 seeds, 4000 steps) and evaluates all four calibration mechanisms post hoc on the same scores and divergence traces, at nominal levels $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$. This design isolates the calibration layer: training is independent of the thresholding mechanism, so differences between mechanisms are not confounded by retraining. Evaluation windows per fold are listed in Table I; labels and protocols per Section III-C.

B. Static Thresholds Fail in Both Directions; ACI Restores Coverage

Table II reports normal-window coverage at $\alpha = 0.10$ (nominal 0.90) and the mean coverage gap across the four nominal levels, averaged over seeds. Static source thresholds fail on every fold, and the failures are not small: on F2 (wind) coverage collapses to 0.18—the source threshold over-alarms normal operation almost constantly—while the other five folds sit at 0.989–0.999, i.e., the threshold is too high and alarms almost never fire. A single source-calibrated number is thus not merely imprecise; it is unusable in practice in opposite directions on different target domains.

Online ACI repairs this on five of six folds: mean gaps of 0.006–0.015 (F1, F2, F4, F5, F6) against 0.077–0.523 for static. Under moving-block bootstrap (seed 0, α=0.10, 400 resamples), the ACI coverage interval stays within ± 0.02 of nominal for F1/F2/F4/F5 at moderate-to-long blocks, while F6 widens with block length (up to $[0.897, 0.933]$ at L=1,000)—consistent with its slow drift—and F3 remains the synthetic exception. Averages are robust to weighting: macro versus window-weighted (micro) mean ACI gaps at α=0.10 are 0.022 vs 0.012 (NA-ACI: 0.021 vs 0.011). The one exception is F3 (solar, synthetic injection): every mechanism stays at coverage ≈ 0.998. This is the alarm-budget effect defined in Section III-A: F3's evaluation windows contain 20% strongly separable injected anomalies, and once ACI has spent the alarm budget on them (87% of injected runs detected, median delay 0), the normal-window FPR is driven to nearly zero. Coverage on F3 therefore over-shoots, and we read F3 as a synthetic-anomaly stress case rather than a calibration success. Figure 2 shows the full α-sweep; the nominal line is tracked on five folds and missed on F3 for the stated reason.

C. The Alarm-Budget Defect and NA-ACI

On anomaly-rich folds, plain ACI spends its alarm budget on anomalies and over-covers normal windows: at α=0.10, F3's normal-window FPR is 0.002 (ACI detects 39% of injected anomalies), F4's anomaly detection rate is only 0.12 while its FPR sits at the nominal 0.09, and F6 detects 19%. Normal-anchored ACI (Section IV-D) reallocates the budget: with κ=3, F4 detection rises to 0.90 at FPR 0.094, F6 to 0.29 at 0.094, F3 to 0.51 at 0.026, and F5 from 0.11 to 0.16 detection at unchanged FPR. Table III summarizes calibration gaps, FPR, detection rates, and range-F_1. Across all folds and

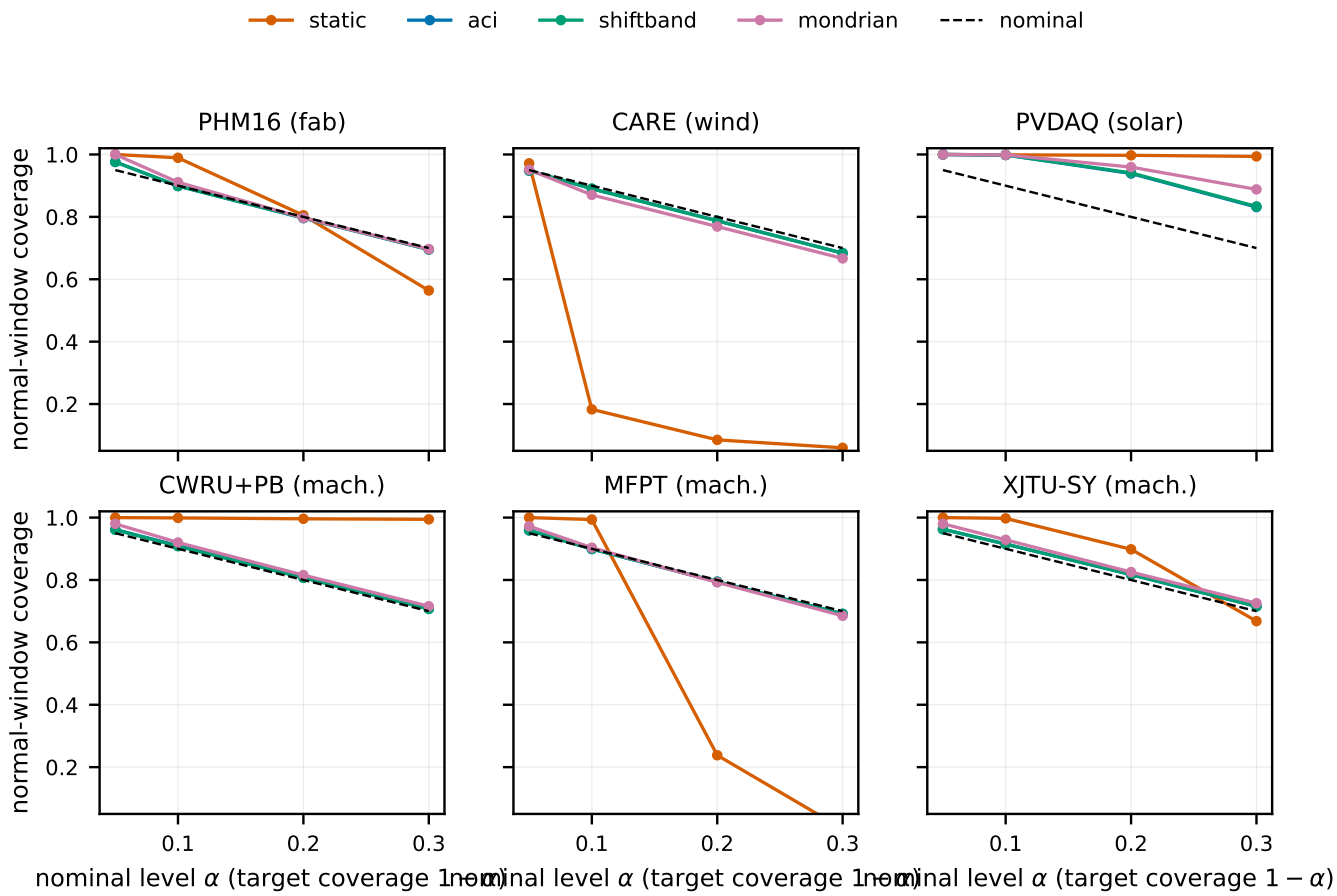


Fig. 2. Normal-window coverage versus nominal level α per fold (mean over seeds). Dashed line: nominal $1-\alpha$. Static thresholds fail in both directions; ACI tracks the nominal level on five folds. F3 (injected synthetic anomalies) over-covers because the alarm budget is consumed by separable anomalies.

TABLE II
NORMAL-WINDOW COVERAGE AT α=0.10 (NOMINAL 0.90, MEAN OF 3 SEEDS), PER-FOLD AUROC AND AUPRC, AND MEAN COVERAGE GAP ACROSS $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$. BEST PER ROW IN BOLD. † BELOW-CHANCE RANKING: EXCLUDED FROM DETECTION-QUALITY CLAIMS. ‡ SYNTHETIC-ONLY FOLD (F3); ITS OVER-COVERAGE STEMS FROM THE ALARM BUDGET BEING CONSUMED BY SEPARABLE INJECTED ANOMALIES. CROSS-VARIANT DIFFERENCES WITHIN SEED SPREAD ARE NOT CLAIMED AS SIGNIFICANT (DESCRIPTIVE COMPARISONS ONLY; THREE SEEDS).

			Coverage @ 0.10				Mean gap	
Fold	AUROC	AUPRC	Static	ACI	Shift	Mondrian	Static	ACI
F1	0.63	0.152	0.989	0.899	0.899	0.911	0.077	**0.009**
F2†	0.39	0.099	0.183	0.890	0.890	0.871	0.523	**0.010**
F3‡	0.79	0.728	0.999	0.998	0.998	0.999	0.160	0.105
F4	0.99	0.992	0.999	0.910	0.908	0.920	0.160	**0.010**
F5†	0.37	0.725	0.994	0.899	0.899	0.904	0.351	**0.006**
F6	0.78	0.682	0.998	0.915	0.915	0.929	0.082	**0.015**

nominal levels, mean range-F_1 improves from 0.265 (ACI) to 0.389 (κ=3; 0.370 at κ=5) while the mean coverage gap stays comparable (0.026 vs. 0.029 at κ=3, and improves to 0.020 at κ=5); anomaly-sparse folds F1/F2 are unchanged at κ=5 and mildly worse at κ=3 (F2 gap $0.010 \rightarrow 0.058$), exposing the expected κ trade-off: aggressive exclusion risks excluding normal windows during scale drift and over-alarms. κ is the method's only new hyper-parameter and can be validated online by monitoring the exclusion rate; we default to κ=5 when anomaly prevalence is low or unknown, κ=3 when prevalence is high.

Temporal diagnosis of the defect. Mean coverage hides the dynamics. On F4 and F6 the ACI threshold traces show a limit cycle: the 5th–95th percentile range of the deployed threshold spans 0.005–2.0 (F4) and 0.014–2.8 (F6)—two to three orders of magnitude—with thousands of oscillation events per stream. The mechanism is integral windup: on anomaly-rich streams the alarm indicator stays on, the relative-gain integrator pushes the threshold up until the stream falls silent, the threshold then decays back, and the cycle repeats—alarm bursts alternate with long blind stretches (F4: longest alarm-free run 491–579 windows, essentially all of it consecutive missed anomalies; F6: $\approx$ 1400-window silences containing 338–346 consecutive missed anomalies). Plain ACI on F4 thus shows FPR 0.090 with detection only 0.118 despite AUROC 0.99: the time-averaged calibration looks nominal while the detector alternates between blindness and bursts, which is how those three numbers coexist. NA-ACI breaks the cycle by construction: with κ=3 the longest consecutive miss streak falls from 491–579 to 51–106 on F4 and from $\approx$ 340 to 36–46 on F6, and detection rises to 0.88–0.92 and 0.29, respectively. We

TABLE III
ALARM-BUDGET DEFECT AND NA-ACI AT α=0.10 (3 SEEDS). MEAN GAP IS OVER FOUR NOMINAL LEVELS. NA-ACI REALLOCATES ALARMS TO ANOMALIES (COLUMNS DET.) AT ESSENTIALLY UNCHANGED NORMAL-WINDOW FPR, WITH LARGE RANGE-F_1 GAINS ON ANOMALY-RICH FOLDS; κ=3 IS AGGRESSIVE, κ=5 CONSERVATIVE.

	ACI		NA-ACI κ=3			NA-ACI κ=5		
Fold	FPR/Det.	R-F_1	FPR/Det.	gap	R-F_1	FPR/Det.	gap	R-F_1
F1	.101/.065	0.07	.105/.066	0.060	0.07	.102/.065	0.012	0.07
F2†	.110/.136	0.13	.137/.159	0.058	0.14	.124/.148	0.031	0.14
F3‡	.002/.389	0.56	.026/.512	0.044	0.63	.019/.506	0.058	0.64
F4	.090/.118	0.19	.094/.895	0.005	0.88	.092/.715	0.006	0.77
F5†	.101/.106	0.19	.101/.158	0.006	0.26	.101/.138	0.006	0.24
F6	.085/.193	0.26	.094/.289	0.004	0.36	.093/.285	0.005	0.36

TABLE IV
EXTREMAL BEHAVIOUR AT α=0.10 (MEAN OF 3 SEEDS, IN WINDOWS): LONGEST ALARM-FREE STRETCH (SIL.), LONGEST CONSECUTIVE MISSED-ANOMALY STRETCH (MISS), AND P95 DETECTION DELAY OVER ANOMALY RUNS. NA-ACI SHORTENS BOTH BLIND EXTREMES; F2/F5 DELAYS ARE NOT DETECTION CLAIMS (BELOW-CHANCE RANKING).

	ACI			NA-ACI κ=3		
Fold	sil.	miss	P95	sil.	miss	P95
F1	116	57	3	116	57	3
F2†	98	20	13	92	20	13
F3‡	65	3	1	58	3	1
F4	549	549	29	455	70	26
F5†	177	177	32	92	84	32
F6	1408	343	309	812	42	6

therefore read the alarm-budget defect as two mechanisms—budget consumption by separable anomalies (F3-type, over-coverage) and windup-induced blindness (F4/F6-type)—both removed by excluding likely-anomalous windows from the feedback path; in control terms, the exclusion acts as anti-windup on the quantile integrator.

Extremal behaviour and decision cost. At α=0.10 the blind stretches of plain ACI coincide with its longest missed-anomaly streaks (Table IV): F4 misses up to 549 consecutive anomalous windows and F6 issues no alarm for 1,408 windows containing 343 consecutive missed anomalies; F6's detection delay is heavy-tailed exactly where it oscillates (P95 = 309 windows versus NA-ACI's 6). NA-ACI cuts the longest miss streak by roughly $8\times$ on both folds ($549 \to 70$; $343 \to 42$). Under a decision-cost view (r = miss-to-false-alarm cost ratio) NA-ACI dominates ACI at every $r \in \{1, 10, 100\}$ (aggregate costs 3,260/27,174/266,314 versus ACI's 4,374/38,789/382,939; static: 3,702/31,101/305,091—competitive only when misses are cheap).

D. Fault-Type Behaviour Under the Windup Defect

F4's aggregate numbers average over mechanically different fault classes. Because the encoder trains on source-normal windows only, every fault class is unseen at score time, and the decomposition below doubles as a held-out-class analysis. Using the released window provenance (Section III-C), we split F4 into its eight CWRU records—class tags follow the standard catalog and are indicative—and the Paderborn late-life windows, and compute per-class AUROC against F4's normal windows and detection rates at α=0.10 (Table V). Two findings. First, the windup blindness is class-blind: all eight CWRU classes rank at AUROC $\geq$ 0.996, yet plain ACI detects them at only 0.04–0.19—the limit cycle, not score quality, decides. Second, NA-ACI recovers seven of the eight records to $\geq$ 0.996 detection; the exception, record 185, recovers only to 0.27, and the Paderborn late-life windows—the only genuinely degraded (non-seeded) class—move from 0.36 to 0.52 at AUROC 0.79. Recovery is thus broad but not uniform: it is weakest where the target score distribution overlaps the threshold region (Paderborn) or where exclusion interacts with the oscillation phase of the walk (record 185). We report both cases rather than hiding them behind the F4 average.

TABLE V
F4 BY FAULT CLASS AT α=0.10 (MEAN OF 3 SEEDS): AUROC AGAINST F4 NORMALS, DETECTION RATE UNDER ACI AND NA-ACI (κ=3). CWRU CLASS TAGS ARE CATALOG-INDICATIVE. NA-ACI RECOVERS SEVEN OF EIGHT CWRU RECORDS; RECORD 185 AND THE PADERBORN LATE-LIFE WINDOWS RECOVER PARTIALLY.

Class	n	AUROC	ACI det.	NA-ACI det.
CWRU-105 (IR007)	907	0.999	0.191	1.000
CWRU-106 (IR014)	906	0.999	0.102	1.000
CWRU-107 (IR021)	915	0.999	0.099	1.000
CWRU-108 (IR028)	919	0.999	0.104	1.000
CWRU-118 (B007)	910	0.998	0.071	1.000
CWRU-130 (OR@6)	926	1.000	0.190	1.000
CWRU-169 (OR@3)	906	0.998	0.037	0.996
CWRU-185 (OR@12)	911	0.996	0.083	0.267
Paderborn late-life	256	0.789	0.363	0.523

E. Calibration Baselines and Update-Rate Ablation

Table VI compares ACI against rolling-quantile tracking, an extreme-value (GPD/POT) static threshold [12], and slower ACI updates at $\alpha = 0.10$ (seed 0; seed trends identical). Rolling-quantile tracking under-covers on every fold (0.851–0.930): without coverage feedback, a windowed empirical quantile cannot keep pace with the shifting score scale. The extreme-value tail fit inherits the static failure exactly, because it is calibrated on source scores. Slower updates degrade progressively—updating every 20th window collapses F2 to 0.699—which justifies the per-window update of Equation 1; we also verified that larger gains trade short-interval stability for no long-run benefit, and we approximate the DtACI-style dual update [10] with these variants.

Table VII validates the quantity ACI is entitled to control: the stream-wide alarm rate. Plain ACI tracks the nominal rate—20 of the 24 fold–level cells deviate by at most 0.015, the largest deviation is 0.033 (F3 at α=0.30)—against static deviations up to 0.70 (F2 at α=0.10: 0.794 vs 0.10). NA-ACI deliberately exceeds the stream rate on anomaly-rich folds (F4: 0.42 at α=0.10) because its alarms go to anomalies; the quantity that stays near nominal for NA-ACI is the normal-window FPR bound $\alpha/(1-\epsilon)$, consistent with the exclusion-rate diagnostic.

TABLE VI
THRESHOLDING BASELINES AT α=0.10 (SEED 0, COVERAGE). EVT/POT IS CALIBRATED ON SOURCE SCORES AND COINCIDES WITH STATIC. ACI UPDATES PER WINDOW; k DENOTES UPDATE EVERY k-TH WINDOW.

Fold	Rolling-q	EVT/POT	ACI k=20	ACI k=5	**ACI**
F1	0.857	0.990	0.973	0.908	**0.897**
F2	0.859	0.185	0.699	0.858	**0.890**
F3	0.930	0.999	0.999	0.999	**0.998**
F4	0.898	0.999	0.997	0.930	**0.910**
F5	0.851	0.995	0.928	0.910	**0.899**
F6	0.909	0.999	0.942	0.934	**0.915**

TABLE VII
STREAM-WIDE ALARM RATE AT α=0.10 (MEAN OF 3 SEEDS). ACI TRACKS THE NOMINAL RATE—THE QUANTITY ITS UPDATE REGULATES—ON EVERY FOLD; STATIC THRESHOLDS ARE OFF BY UP TO 7×; NA-ACI SPENDS ADDITIONAL ALARMS ON ANOMALIES BY DESIGN.

Fold	Static	ACI	NA-ACI κ=3	NA-ACI κ=5
F1	0.011	0.097	0.100	0.098
F2	0.794	0.113	0.139	0.126
F3	0.067	0.079	0.123	0.116
F4	0.273	0.102	0.421	0.346
F5	0.067	0.105	0.143	0.128
F6	0.078	0.111	0.140	0.139

F. Target-Side Calibration Baselines

We also test whether purely *target-side* calibration can replace the feedback loop: (i) a fixed quantile of the first N=512 window scores; (ii) an expanding-window quantile; (iii) periodic recalibration every 5,000 windows; and (iv) a DSPOT-style streaming POT that refits a GPD tail over a sliding 2,048-window buffer every 64 steps. Table VIII reports coverage gaps at α=0.10. No target-side baseline brings the gap inside 0.02 on the difficult folds. On F4 the opening-window quantile is catastrophic (0.890: the first windows already contain 41% anomalies, poisoning the quantile into 45% alarms) while expanding and periodic quantiles stay at 0.37/0.65 and DSPOT—the strongest target-side rule—reaches only 0.094, an order of magnitude above ACI (0.010). On F6 the opening quantile overshoots the source scale 317× and nearly silences the stream (0.007 alarm rate), and DSPOT stays at 0.164 (ACI: 0.015). F1 is the one exception—a naive expanding quantile matches ACI (0.005 vs 0.009)—consistent with its slow drift being trackable without feedback. On F2–F6 the best target-side rule still trails ACI by roughly 3–11×. The feedback loop is thus what carries the difficult folds; NA-ACI changes how the feedback budget is spent, not whether feedback is needed.

G. Sensitivity, Tracking, and Warm-Up

κ and γ. Sweeping $\kappa \in \{1.5, 2, 3, 5, 10\}$ confirms the frontier. On F4, κ=2–3 maximizes detection (0.95/0.89) at gaps 0.008/0.005; on the anomaly-sparse folds small κ over-excludes and degrades coverage (F1: gap 0.198 at κ=1.5 vs 0.012 at κ=5, approaching ACI's 0.009), which grounds the default rule of Section V-C: large κ when prevalence is unknown, small when it is high. Sweeping the gain, ACI's mean gap is minimized at γ=0.05–0.10 (0.026/0.021) with the F4 oscillation count growing monotonically (2,540 → 3,073 as γ goes 0.01 → 0.10); NA-ACI degrades at large gains (0.029 → 0.061 from γ=0.05 to 0.20). We keep γ=0.05.

TABLE VIII
TARGET-SIDE CALIBRATION BASELINES AT α=0.10 (COVERAGE GAP, MEAN OF 3 SEEDS): FIXED FIRST-512 TARGET QUANTILE, EXPANDING-WINDOW QUANTILE, PERIODIC RECALIBRATION (K=5,000), DSPOT-STYLE STREAMING POT. COMPARE ACI BY ROW (TABLE II); ONLY F1-EXPANDING IS COMPETITIVE.

Fold	first-N	expanding	periodic	DSPOT
F1	0.017	**0.005**	0.017	0.021
F2	0.039	0.039	0.039	0.036
F3‡	0.350	0.361	0.350	0.321
F4	0.890	0.370	0.649	0.094
F5†	0.060	0.061	0.065	0.026
F6	0.070	0.226	0.202	0.164

Pure tracking is not calibration. A P^2 streaming quantile tracker—online but without feedback—fails on every fold (coverage gaps 0.10–0.88): tracking the score distribution misplaces the threshold whenever the distribution drifts or carries anomalies, isolating the value of the feedback path itself (cf. the target-side baselines above).

Warm-up. The first window after which ACI's running normal-window coverage stays within ±0.02 of nominal forever is fold-dependent: ≈380–400 (F5/F1), ≈1,800 (F2), ≈2,500 (F4), ≈11,300 (F6), and never on the synthetic fold F3. F6's 11k-window warm-up (of 16.8k) mirrors its 317× initial threshold ratio: cold-starting far from the source is operationally expensive, and warm-up-safe initialization is a deployment prerequisite.

Per-window α versus per-hour budgets. The window step spans five orders of magnitude across folds (128 s at 1 Hz versus 1–10.7 ms at kHz rates), so the same α means very different operational rates: at α=0.10 the kHz folds issue 7×10^4–3.6×10^5 alarms per hour per stream, and a one-per-hour budget would require $\alpha \approx 10^{-7}$–10^{-6}—below what a 1.7×10^4-window evaluation can statistically resolve (less than one expected alarm); the slow folds sit at the opposite extreme (≈ 0.004 alarms/h at α=0.10). Per-window coverage and per-hour alarm budgets are therefore not interchangeable: deployments should report the window step alongside α, and fast streams need either a resolution floor or longer evaluation streams.

Mechanism alternatives. A compact comparison (mean gap at α=0.10 / F4 gap / F6 gap): a fixed-step additive update (γq_{src}) reaches 0.031/0.011/0.033; a DtACI-style mixed update 0.024/0.008/0.019; PID-style updates with proportional and derivative terms 0.023/0.010/0.015, trimming the F4 oscillation count only modestly (2,992 → 2,865). None approaches NA-ACI's F4/F6 gaps (0.006/0.006): the windup defect is fixed by restricting the feedback path, not by adding controller terms. Weighted conformal prediction—score-space density-ratio reweighting of the source-calibration scores [15]—

reaches near-nominal coverage on F2 (0.002) but degrades on F6 (0.100) and F3 (0.098): one-pass reweighting cannot track an online scale drift whose support barely overlaps the source. An automatic fallback (monitor the alarm rate; revert to plain ACI when it leaves $[0.5\alpha, 2\alpha]$) barely moves F2 ($0.037 \to 0.034$) and discards NA-ACI's F6 gain ($0.006 \to 0.015$); we therefore keep the exclusion-rate monitor as a passive diagnostic rather than an automatic mode switch. Estimating prevalence from the score distribution to set κ automatically also fails—a two-component fit on log-scores estimates F1 prevalence at 0.82 against ground truth 0.12—so κ remains a monitored user choice.

Incipient faults. Replacing each fold's longest anomaly run with a slow score ramp (from $0.7\times$ to $\{1.3, 1.6, 2.0\}\times$ the source quantile over runs of 100–7,300 windows) probes the failure mode the exclusion rule risks: every ramp window falls below κq_t and enters the feedback path. The measured penalty is small: NA-ACI delays detection by at most 6 windows relative to ACI (F6: $20 \to 26$ windows at the slowest ramp; identical on the other folds), while ramps are intrinsically harder than step anomalies for both mechanisms (F6 baseline delay 1 window versus 19–26 on ramps). Ramps slower than these remain untested—a stated boundary of this score-space stress test; raw-signal incipient-fault injection needs a dedicated dataset and is left to future work.

Operational filtering and cost. Industrial alarm stacks debounce with m-of-n voting; on these streams it fails structurally: requiring $m{=}3$ consecutive threshold exceedances collapses the alarm rate by 20–$50\times$ and detection to near zero (F1: $0.066 \to 0.001$), because exceedances are noncontiguous spikes rather than persistent states—event-level aggregation (one alert per alarm run), not window-level voting, is the compatible filter; hysteresis bands are approximately neutral (coverage ± 0.01, detection ± 0.007). For deployment cost we exported the F4 encoder to ONNX (numeric equivalence within 3×10^{-9} after replacing the FFT band selector with an equivalent fixed-basis matmul) and benchmarked it single-threaded on a desktop CPU: 0.55 ms/window at batch 1 and $91\,\mu$s/window at batch 64 (PyTorch CPU: 1.1 ms/window), with the calibration update adding $0.36\,\mu$s. INT8 dynamic quantization shrinks the model from 0.46 to 0.30 MB and leaves the calibration outcome essentially unchanged (Spearman 0.999 against FP32 scores; coverage gap $0.0097 \to 0.0095$; detection $0.118 \to 0.118$). A checkpointed streaming implementation (per-window scoring $\to$ ACI update $\to$ alarm, with state save/restore verified identical after resume) runs end-to-end at 0.61 ms/window single-threaded on this host; Raspberry-Pi-class and MCU deployments require the corresponding hardware and remain future work. Across seeds, gap spread is at most 0.005 and AUROC spread at most 0.02, so three seeds bound the seed-level variability of every claim we make; cross-variant differences within seed spread are stated as no-difference rather than significant, and all cross-variant comparisons are descriptive (no formal multiplicity-corrected significance claims).

H. Divergence-Shifted and Conditional Variants: No Gain

Both target-aware refinements fail to improve on plain ACI. The divergence-shifted threshold (Section IV) is statistically indistinguishable from ACI on every fold and nominal level (coverage differences ≤ 0.002, far below the seed spread), and we therefore report no added value rather than equivalence: the ACI feedback term itself absorbs the score-scale change the divergence term was meant to anticipate. Mondrian ACI shows no improvement either; its point-estimate mean gaps are up to roughly twice ACI's (0.012–0.124 vs 0.006–0.105), and we note that this comparison is descriptive—with three seeds per fold we do not claim significance on the small differences. Splitting the calibration stream into divergence regimes shrinks each regime's feedback sample more than the conditional targeting gains. Figure 2 includes both variants; we report them as negative controls rather than contributions.

I. Transfer Cost Is Real but Not Predicted by Divergence

Keeping coverage across domains is not free. Mean threshold inflation at $\alpha = 0.10$ spans two orders of magnitude across folds: $0.5\times$–$0.6\times$ (F1, F3), $1.3\times$ (F5), 2.9–$3.1\times$ (F4), 10.5–$11.0\times$ (F2), and 17.8–$19.2\times$ (F6)—the XJTU-SY target needs a threshold roughly twenty times the source quantile to hold its coverage. An oracle decomposition separates the two effects this number conflates: the $(1-\alpha)$ quantile of the target's *true* normal scores (labels used offline, for diagnosis only) sits at $0.07\times$ the source quantile on F4 and $0.62\times$ on F6, while ACI's mean deployed threshold sits $9.4\times$ *above* that oracle on F4—a windup over-shoot larger than the domain gap it responds to—and at $0.71\times$ of it on F6 (under-tracking). Inflation alone mixes over-shoot with under-shoot. Grouping folds by shift type—cross-family targets (F1–F3: CMP, wind, PV) versus machinery-heavy mixtures (F4–F6)—the conclusions hold in both subsets (≤ 0.105 / ≤ 0.015 ACI gaps respectively), with NA-ACI's $\kappa{=}3$ cost landing in the sparse cross-family folds (F1/F2: 0.060/0.058 vs ACI's 0.009/0.010) and its gains in the same-family folds (F4: 0.005, F6: 0.004).

Figure 4 shows the inflation bars and exemplar threshold traces: on F2 the static threshold sits orders of magnitude below the target score scale (hence the 0.18 coverage), while ACI ramps q_t into the target regime; on F6 the trace climbs steadily as the quantile tracks the shifting scores.

The divergence hypothesis, however, does not survive: across folds and seeds, mean band divergence $\overline{D}^{\text{band}}$ and inflation show no reliable association, with Pearson $r \in \{0.13, 0.19, 0.26, 0.30\}$ for $\alpha \in \{0.10, 0.05, 0.20, 0.30\}$ ($n{=}18$ per level, all $p > 0.22$). An earlier smaller-sample estimate ($n{=}8$) had suggested a borderline correlation; with three seeds it vanishes. Given the small sample, we treat this as evidence against a strong predictive claim rather than a precise null estimate: we keep inflation as a descriptive transfer-cost diagnostic and drop the claim that divergence anticipates it.

J. Calibration and Detection Quality Decouple

AUROC per fold spans the full unusable-to-strong range: 0.63 (F1), 0.39 (F2), 0.79 (F3), 0.99 (F4), 0.35–0.40 (F5),

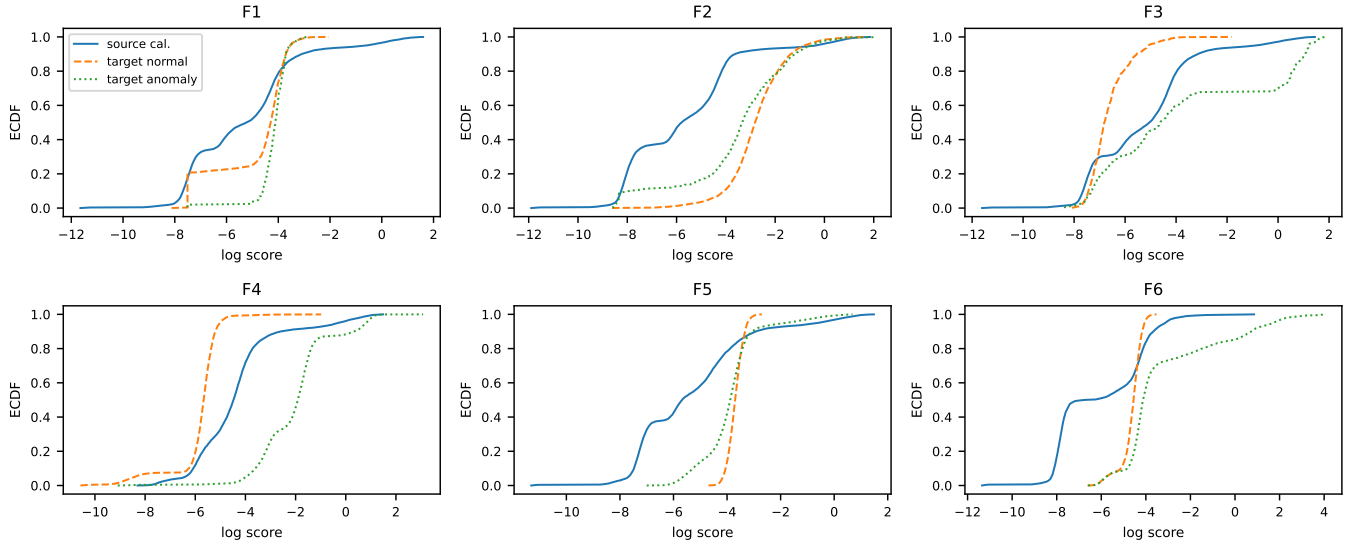


Fig. 3. Source-calibration versus target score distributions (log-score ECDFs, seed 0). The inflation numbers have shapes behind them: F2 is a large upward shift with substantial support overlap; F4's target scores sit far below the source-calibration scale (near-disjoint), the regime where static thresholds under-alarm and the relative-gain controller winds up.

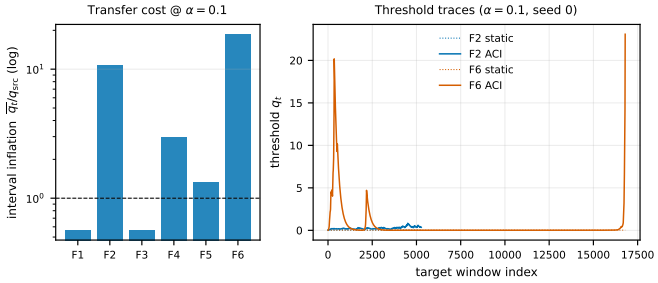


Fig. 4. Left: threshold inflation $\overline{q_t}/q_{\mathrm{src}}$ per fold (α=0.10, log scale). Right: threshold traces on F2 and F6 (static vs ACI, seed 0). Static thresholds are mis-scaled in opposite directions; ACI tracks the target regime.

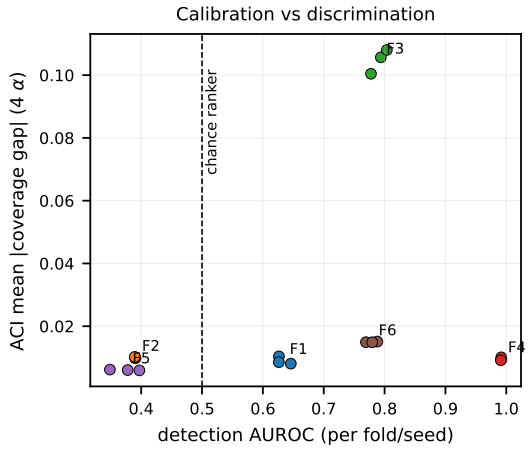


Fig. 5. Calibration versus discrimination: ACI mean coverage gap (four nominal levels) against AUROC for every fold and seed. Below-chance folds (F2, F5) remain coverage-controllable, but their alarms carry no detection utility—an explicit caveat, not a success.

0.78 (F6), with per-seed spread below 0.02. Two folds rank anomalies *below* chance. We diagnosed both: on F2 and F5, simple feature baselines—window RMS, low-band and high-band energies—also score below 0.5 (F2: 0.41/0.45/0.37; F5: 0.36/0.39/0.33), while the F6 control scores 0.72–0.82 on the same features. The inversion is a property of these datasets under their labeling protocols (fault-labeled periods carry systematically lower amplitude/spectral statistics in this windowing), not a failure unique to our encoder; no repair within our protocol is available, and we therefore **exclude F2 and F5 from detection-quality claims**. Figure 5 makes the decoupling explicit: ACI's coverage gap stays near nominal even at AUROC < 0.5—coverage control is a statement about the threshold, not about the usefulness of the alarms. **Calibration cannot rescue an anti-correlated ranker**; deployments must audit ranking first (our feature-baseline check is a two-line diagnostic) and only then trust calibrated coverage as a reliability statement.

Detection-side numbers, reported for the four trustworthy folds, are consistent with the coverage picture: false-alarm rates at α=0.10 sit at 85–110 per 1000 normal windows on F1/F2/F4/F5/F6 (F3: 2.3), and detection delays on F1/F4/F6 are 1–2 windows—roughly 2–4 min on F1 at 1 Hz, 2–21 ms on F4 at 12/128 kHz, and 5–10 ms on F6 at 25.6 kHz—while F5's single run is detected after 49 windows ($\approx$ 0.13 s at the fault-section rate). Window-level AUPRC echoes the ranking picture: it falls below the anomaly prevalence on exactly the two below-chance folds (F2: 0.099 vs 0.106; F5: 0.725 vs 0.729, where the 73% prevalence dominates the absolute value), while F4 reaches 0.992. Range-F_1 follows the overlapping-run convention (an alarm is credited to a run if it intersects it); the implementation ships with the code. The synthetic F3 study is stable under contamination: AUROC stays within 0.73–0.83 for injection fractions $\{0.05, 0.10, 0.20, 0.40\}$ across seeds.

K. Contaminated Initialization and Label-Noise Robustness

Two deployment-side robustness checks (using the first 512 evaluation windows as the deployment calibration stream). *Contaminated initialization*: if the threshold is initialized from a target segment that contains anomalies, the frozen target-quantile rule degrades exactly as feared—on F4, injecting 10% anomalies into the initialization segment pushes the quantile from 0.07 to 0.12 of the source scale (still far too low; the frozen gap stays at 0.89)—whereas ACI started from the same contaminated value re-converges online: its stream-level gap is 0.018 on F4, 0.001 on F2, and 0.006 on F5, essentially independent of contamination level, at the cost of a slightly longer warm-up (F6's first-1,000-window gap rises from 0.048 to 0.067 at 10% contamination; F3, whose evaluation window is shortest at 1,072 windows, converges slowest). *Label noise*: neither ACI nor NA-ACI uses label information, so symmetric label flips affect evaluation metrics only. At 5/10/20% flip rates the ACI coverage gap stays at 0.022/0.021/0.018, the NA-ACI gap moves from 0.024 to 0.039, and static thresholds remain unusable (0.17–0.19); ranking quality degrades gracefully (mean AUROC $0.64 \rightarrow 0.58$). Calibration reliability under noisy supervision thus follows the decoupling caveat of Section V-J: the threshold controller is label-free and stays calibrated; only our ability to *measure* detection quality depends on labels.

L. Calibration Conclusions Are Not Encoder-Specific

To test whether the calibration findings depend on our neural encoder, we repeated the comparison on folds F1/F4/F6 with two non-neural score producers: PCA reconstruction (32 components on flattened windows) and a Mahalanobis distance on 16-dimensional band-energy features. Table X reports mean coverage gaps across the four nominal levels. Static calibration fails for these producers as well—including both failure directions *within* the feature-based producer (F4 over-alarms at coverage 0.06; F6 never alarms at 0.000)—while ACI restores gaps to at most 0.011 in all six producer–fold combinations, even when the producer's ranking is below chance (F1 Mahalanobis, AUROC 0.47). The calibration layer is agnostic to score provenance, and the decoupling caveat of Section V-J applies equally to classical detectors.

We further evaluate a *zero-shot time-series foundation model* as an external score producer: Chronos-bolt-small [5], scored by its 32-step forecast residuals (context 224; median quantile; 4,000 stratified evaluation windows per fold, same splits and injection protocol). Table IX reports the outcome. The pattern replicates in full: static calibration fails on every fold and in both directions (mean gaps 0.066–0.571); ACI restores gaps to 0.009–0.048; NA-ACI improves further where the budget defect bites (F3: $0.048 \rightarrow 0.035$, F6: $0.028 \rightarrow 0.014$; F4 range-F_1 $0.27 \rightarrow 0.39$ with κ=3). Two findings stand out. First, Chronos-bolt ranks anomalies *below chance on the same two folds* as our encoder (F2 AUROC 0.34, F5 0.32), which strongly supports reading the inversion as a dataset/label property rather than an encoder artifact. Second, our 78K-parameter reconstruction encoder matches or exceeds the zero-shot foundation model on AUROC in all six folds (e.g., F3 0.79 vs. 0.65, F4 0.99 vs. 0.94), consistent with recent evidence that generic foundation models are not automatically superior on shifted industrial series [7], [8].

TABLE IX
ZERO-SHOT CHRONOS-BOLT-SMALL EXTERNAL SCORE PRODUCER (FORECAST-RESIDUAL SCORES; 4,000 STRATIFIED WINDOWS PER FOLD). STATIC FAILS BOTH WAYS; ACI RESTORES COVERAGE; NA-ACI TRACKS THE BUDGET DEFECT. AUROC BELOW 0.5 ON F2/F5 MATCHES OUR ENCODER—A DATA/LABEL PROPERTY, NOT AN ENCODER ARTIFACT.

Fold	AUROC	Static gap	ACI gap	NA-ACI gap	R-F_1 ACI	R-F_1 NA
F1	0.63	0.124	0.032	0.037 (κ=5)	0.09	0.10
F2	0.34	0.571	0.015	0.041 (κ=5)	0.13	0.13
F3	0.65	0.139	0.048	0.035 (κ=5)	0.30	0.30
F4	0.94	0.066	0.009	0.006 (κ=5)	0.27	0.39
F5	0.32	0.324	0.012	0.012 (κ=5)	0.25	0.28
F6	0.74	0.098	0.028	0.014 (κ=5)	0.37	0.40

TABLE X
MEAN COVERAGE GAP OVER FOUR NOMINAL LEVELS FOR NON-NEURAL SCORE PRODUCERS (STATIC VS. ACI). ACI RESTORES COVERAGE IN EVERY COMBINATION; STATIC FAILS, INCLUDING IN BOTH DIRECTIONS WITHIN THE MAHALANOBIS PRODUCER.

Producer	F1	F4	F6
PCA reconstruction (static / ACI)	0.154 / **0.002**	0.153 / **0.004**	0.163 / **0.011**
Mahalanobis band (static / ACI)	0.546 / **0.004**	0.798 / **0.002**	0.838 / **0.003**

M. Comparison with Rival Cross-Domain Detectors (Mechanism-Level)

Because the official codebases of CICADA [3] and DACAD [4] target different benchmarks, we reimplement their core mechanisms under our six-fold protocol and label the comparison accordingly: a *DACAD-style* scorer (anomaly injection + discriminative head + source–target moment alignment) and a *CICADA-style* scorer (four reconstruction experts over source groups with divergence-based expert weighting). Table XI reports ranking quality and calibration. Three observations. First, the calibration result is universal: static thresholds fail for both rivals on every fold (mean gaps 0.037–0.499, including DACAD-style F2 at 0.037 and CICADA-style F2 at 0.499), ACI restores them to 0.001–0.101, and NA-ACI helps where the budget defect bites (CICADA-style F3 gap $0.101 \rightarrow 0.059$; F4 range-F_1 $0.24 \rightarrow 0.72$). Second, ranking quality of the rivals is unstable across domains—DACAD-style ranges from AUROC 0.26 (F4) to 0.85 (F3), and CICADA-style from 0.32 (F5) to 0.99 (F4)—so neither is a reliable default. Third, no scorer dominates: our plain encoder matches the best rival on F1/F4/F6 and trails on F2/F3/F5, while every scorer's ranking swings across folds (ours 0.37–0.99, DACAD-style 0.26–0.85, CICADA-style 0.32–0.99)—supporting calibration as the first-order problem for deployment, with ranking left fold-dependent.

N. Runtime, Memory, and Negative Controls

The calibration layer is cheap: across all folds, training takes $\sim$70 s, target scoring at most 0.41 s, and one ACI update costs 0.6–$0.8\,\mu$s, with 26 MB peak GPU memory (Table XII). The

TABLE XI
MECHANISM-LEVEL RIVAL COMPARISON (6 FOLDS). AUROC PER SCORER; MEAN COVERAGE GAP STATIC → ACI; RANGE-F_1 (MEAN OVER LEVELS) FOR ACI AND NA-ACI. STATIC FAILS FOR BOTH RIVALS; ACI RESTORES; RIVAL RANKINGS ARE UNSTABLE ACROSS FOLDS. BEST PER-ROW AUROC IN BOLD.

	AUROC			Gap (static→ACI)		R-F_1
Fold	Ours	DACAD*	CICADA*	DACAD*	CICADA*	NA gain
F1	**0.63**	0.60	0.62	.065→.003	.075→.006	—
F2	0.39	**0.49**	0.35	.037→.003	.499→.010	—
F3	0.79	**0.85**	0.84	.155→.093	.155→.101	0.59 (k5)
F4	0.99	0.26	**0.99**	.064→.002	.158→.006	0.24→0.72
F5	0.37	**0.59**	0.32	.076→.002	.238→.004	0.26→0.27
F6	0.78	0.53	**0.78**	.143→.001	.133→.011	0.26→0.34

TABLE XII
PER-RUN COST (ONE FOLD, ONE SEED; RTX 3060 TI). ACI UPDATE LATENCY MEASURED OVER ALL EVALUATION WINDOWS AND NOMINAL LEVELS.

	Train (s)	Score (s)	Update (μs)	Peak mem (MB)	Params
Value	$\sim$70	$\leq$0.41	0.6–0.8	26	$\sim$78 K

thresholding mechanisms differ negligibly in cost. For completeness we record the negative controls from the predecessor study: spectral-invariance, relation-consistency, and band-saliency losses, low-rank test-time adaptation, and divergence-based source gating each changed range-F_1 by at most 2.2% across four folds, and encoder identity (plain vs. regularized vs. gated) moves calibration coverage by at most 0.002—which is why this paper studies the calibration layer rather than detector architecture.

VI. DISCUSSION AND DEPLOYMENT GUIDELINE

A. What We Would Tell a Practitioner

Four rules follow directly from the study. (1) *Never ship a source-calibrated static threshold for a new domain.* Across four domain families the static threshold failed on six of six folds and in both directions; the failure is not a tuning problem but a distributional one. (2) *Wrap the score with an online ACI layer and monitor its trace; use NA-ACI when anomalies may be abundant.* On five folds the layer recovered coverage within 0.015 of nominal at negligible cost (0.6–0.8 μs per decision); on anomaly-rich streams, NA-ACI keeps the budget on the normal regime and improves detection sharply at essentially unchanged FPR, at the price of one interpretable parameter κ whose exclusion rate is monitorable online. (3) *Report coverage and alarm rate separately, and know which one is guaranteed.* ACI controls the long-run alarm rate; normal-window FPR control follows when anomalies are rare relative to α, and NA-ACI narrows the gap otherwise. (4) *Audit the ranker before trusting the calibration.* Two folds here rank anomalies below chance; their coverage numbers look excellent and their alarms are worthless. A feature-baseline AUROC check (RMS or spectral energy) costs nothing and exposes this. A monitoring protocol closes the loop: (1) audit ranking with a feature-baseline AUROC, and stop if it is below chance; (2) run ACI by default, NA-ACI when prevalence is believed high; (3) watch three quantities—alarm-rate EWMA (band $[0.5\alpha, 2\alpha]$), exclusion rate (compare against the prevalence estimate), and threshold drift (ADWIN/Page–Hinkley [16], [17])—with out-of-band signals triggering human inspection and persistent drift triggering recalibration; (4) recalibrate from a fresh labeled audit segment whenever the ranking audit fails.

B. A Measurement Perspective

Read through a metrology lens, the deployed threshold is a *measured quantity*—the output of the calibration procedure—and this paper quantifies its transfer behavior: static calibration carries a transfer error above 0.7 coverage on the worst fold, while the online layer holds it to an $O(\gamma)$ tracking error with an empirical seed spread below 0.02. In VIM/GUM terms, the oracle decomposition of Section V-I separates the *domain term* (how far the target scale has moved) from the *mechanism term* (how far the controller sits from the target-optimal threshold), and the coverage/alarm-rate pair is the accuracy/precision-style decomposition of the calibration, reported separately by construction. We stop short of a full uncertainty budget: threshold uncertainty is not propagated into event-level false-alarm probabilities, which is the natural next step for a measurement-audience treatment.

C. Limitations

The benchmark has six folds over four domain families, with single-dataset wind, one fab dataset, a photovoltaic family represented only by its synthetic-injection fold, and three of six folds targeting rotating machinery; "cross-domain" here means cross-population within industrial monitoring, not six separate industries. Pseudo-labels are unavoidable for several datasets, and PHM16's q_{10} removal-rate protocol and the PVDAQ synthetic injection are the weakest links; both are documented and their sensitivity is reported (contamination sweep, diagnosis of below-chance folds). The encoder family is deliberately simple and uniform; we compare against mechanism-level reimplementations of the closest rivals (CICADA-, DACAD-style, Section V-M) and against classical and foundation-model score producers (PCA, band-energy Mahalanobis, Chronos-bolt-small), while official-code comparisons remain a revision item. Finally, the conditional-coverage goal remains open: Mondrian and divergence-shifted variants did not deliver it, and NA-ACI narrows but does not close the conditional gap, so we claim no more than marginal coverage control with an improved budget allocation. Equation 1 is a scale-free variant of ACI; its long-run alarm-rate interpretation is $O(\gamma)$ in the log-threshold parameterization and does not inherit the exact finite-sample constants proved for the additive form. The source pools are also imbalanced (1,024–16,007 normal windows per dataset after capping, a $15.6\times$ spread); balanced-domain retraining remains untested. Dataset licences differ by provider (Paderborn via the KAt DataCenter terms, PVDAQ under NREL's terms, NASA and university datasets under research-use terms); we redistribute no raw data and defer licence specifics to each provider. The rival comparison is mechanism-level: our reimplementations preserve each rival's core mechanism (injection plus alignment head; expert mixture

with divergence gating) under identical windows and training budgets, and official-code comparisons remain future work with the alignment protocol documented in the released repository.

D. Future Work

Three directions follow. First, conditional coverage beyond marginal control: NA-ACI fixes the allocation of the alarm budget but does not provide group-conditional guarantees, and the assumption behind its exclusion rule (anomaly-pure tail) deserves theory that links measurable diagnostics—exclusion rate, score-distribution drift—to coverage bounds. Second, extending the benchmark with more domain families and externally produced scores (foundation models, domain-adaptation detectors) to test whether the calibration conclusions and NA-ACI gains transfer to other score producers. Third, a theory-side account of threshold inflation: inflation spans two orders of magnitude across our folds and we can describe it but not yet predict it.

VII. Conclusion

Deployed anomaly detectors are thresholds, not scores, and a threshold calibrated on one domain breaks on the next. Across six leave-one-domain-out folds over four domain families of industrial time series, static source-calibrated thresholds failed in both directions—over-alarming to 0.18 coverage on wind data and under-alarming to 0.999 elsewhere—while an online adaptive-conformal layer restored normal-window coverage to within 0.015 of nominal on five folds at 0.6–$0.8\,\mu$s per decision. Target-aware refinements (divergence-shifted, conditional) brought no gain, divergence did not predict the $0.5\times$–$19\times$ transfer cost, and on two folds whose scores rank anomalies below chance, calibration controlled coverage but could not make the alarms useful. We release the benchmark, protocols, and code at https://github.com/JibiaoZhu/crossdomain-conformal-calibration (archived at https://doi.org/10.5281/zenodo.23321794), and we state the negative results plainly: deploy calibration, audit the ranker first, and never trust a source threshold on a new domain.

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